

Labreport - Machine learning models for human activity classification using smartphone sensors

Linn Hamadé ¹ *

¹ HKR department of computer science; linn.hamade0284@stud.hkr.se

Abstract: Smartphone sensors has the ability to capture motion data that can capture patterns of human activities such as walking, running, jumping and squats. This was done by the use of the built in accelerometer and gyroscope. Readings of the motion data was processed and divided into training, Validation and test sets. The data split allowed the performance of different classifiers to be compared. Classifications algorithms including K-Nearest Neighbors, normalized KNN, decision trees and Neural Networks featuring two hidden layers with either 10 or 50 neurons, tested both with and without feature normalization, and lastly recurrent neural network based on gated recurrent units was implemented. The Models performance was measured through accuracy and confusion matrices, and compared against each other to determine which achieved the best performance. Results showed that the GRU model, consisting of one bidirectional layer with 64 neurons, a window length of 50 samples, a dropout rate of 0.3 and a learning rate of 0.001 achieved the highest test accuracy. While the GRU model consisting of one bidirectional layer with 128 neurons, a window length of 25 samples, a dropout rate of 0.2 and a learning rate of 0.001 achieved the best validation accuracy, outperforming all other classification models. This shows that data preprocessing and network design have a impact on classification performance. The results indicate that recurrent neural network based on gated recurrent units can capture activity patterns from raw sensor data. A separate 90-second recording containing all movements was used to evaluate the model's performance on continuous data. The model performed well on the original participant's recording but showed reduced accuracy on the partner's data, indicating limited generalization to unseen subjects.

Keywords: smartphone sensors; accelerometer; gyroscope; neural networks; K-Nearest Neighbors; decision tree; Gated recurrent units

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1. Introduction

Human activity recognition is an important field in machine learning that focuses on identifying human movements from sensor data. With the growing use of smartphones and wearable devices equipped with motion sensors, such as accelerometers and gyroscopes, it has become possible to analyze and classify different physical activities in everyday settings. This study aims to explore how motion data collected from a smartphone can be used to classify common activities using different machine learning algorithms and to evaluate their performance in activity recognition.

2. Materials and Methods

The data used in this study were collected using a smartphone placed, as illustrated in Figure 1. The phone's built in accelerometer and gyroscope sensors recorded motion data

corresponding to four different activities: walking, running, jumping and squats. Each recording generated six sensor readings: acceleration and angular velocity along the x, y and z axes.



Figure 1. Position of the smartphone in the pocket during data collection.

The dataset was divided into training, validation and test sets. All sets included both accelerometer and gyroscope data. Since the raw signals often contained noise at the start and end of each recordings, a trimming function was applied to remove a fixed number of samples from both ends. Missing values were replaced with column means. The cleaned datasets were stored as pickle files. A summary statistics and boxplots was as well generated to identify potential outliers.

The cleaned data was classified using, supervised learning algorithms which are classical machine learning algorithms, including K-Nearest Neighbors (KNN), a normalized version of KNN, a Decision tree classifier, and deep learning algorithms which are Neural Network and gated recurrent units.

KNN was chosen because it performs well on small datasets. Unlike KNN, which makes predictions directly from the stored training data without building a model, the Decision tree creates a model that is easy to understand, compare and visualize. Normalized KNN was used to ensure all features contribute equally to distance calculations. While Neural Networks were included to capture complex, non-linear patterns, with normalization to help stabilize and speed up the training. Finally the GRU (Gated Recurrent Unit) was chosen because it is specifically designed to handle sequential data, such as time series from accelerometer and gyroscope sensors, so it capture patterns in sequential data efficiently.

For the KNN model the acceleration and angular velocity along the x, y and z axes was the inputs and the activity type was the output class. A grid search was conducted over (k) values ranging from 1 to 20 using GridSearchCV with five fold stratified cross-validation. The optimal number of neighbors (k) was selected based on the highest cross-validation accuracy. The final KNN model was trained on the complete training set and evaluated on both validation and test datasets. The classification accuracy and a confusion matrix was also provided. A normalized version of KNN was also used scaling all features to the range $[-1, 1]$. The scaler was fit only on the training data. The same grid search configuration was used to identify the optimal (k) value. The accuracy and confusion matrix were computed and compared with the regular KNN model.

While for the decision Tree classifier, two hyperparameters were implemented using GridSearchCV, the tree depth max_depth ranging from 1 to 20, and the splitting criterion, evaluated for both gini and entropy. The depth range (1–20) was selected to balance model complexity and generalization, while the different splitting criteria were tested to evaluate their effect on classification accuracy. Five fold stratified cross-validation was also used for the decision Tree. After determining the best hyperparameter combination, two models one using the Gini index and one using entropy, were trained and compared to analyze

the impact of the criterion on accuracy. The final model was trained using the optimal parameters and evaluated on both validation and test sets. The resulting confusion matrix was plotted, and the tree structure was also given.

A Normalized and not normalized Neural Network was developed using TensorFlow and Keras with six input neurons representing the sensor features. Two configurations were tested, one with 10 neurons per hidden layer using early stopping (patience = 8) and another with 50 neurons per layer combined with a dropout rate of 0.3. Both methods were applied to reduce overfitting. The hidden layers used ReLU activation, and the output layer applied a softmax function to classify the four activity types. Training was performed for 50 epochs with a batch size of 32. Training and validation accuracy and loss were plotted, which was evaluated on the test set using accuracy and a confusion matrix.

In the final stage of the work a recurrent neural network (RNN) using gated recurrent units (GRU) was implemented. This work was done in three stages (grade 3-5). For grade 3, a GRU model was implemented to classify four movement classes (jumping, running, walking, and squat) using the same accelerometer and gyroscope recordings as the other classification models. The data were divided into overlapping time windows of 25, 50, and 100 samples. Different network configurations were tested by varying the number of layers, neurons, and dropout rate. The model was trained using separate training, validation and test sets. The final model was applied to a 90-second combined recording of all four movements to visualize predicted activities over time.

For grade 4, the hyperparameter search was expanded to include bidirectional GRU layers and variation in the learning rate. Training was performed using 3 fold cross validation. The model was again trained on the four movement classes, and the best configuration was evaluated on the test data and applied to the 90-second combined recording for performance visualization.

Finally for grade 5, the approach was extended to classify both movement and person, resulting in eight distinct classes. The model used combined datasets from two participants and maintained the same GRU based structure, training setup and windowing method. After training and evaluation, the model was used to predict and visualize the activity sequences for both participants 90-second recordings.

3. Results

As presented in Table 1, the best performance was achieved by the deep learning GRU model used in Grade 5. The model consisted of one bidirectional GRU layer with 64 neurons, a window length of 50 samples, a dropout rate of 0.3, and a learning rate of 0.001. It reached a training accuracy of 0.9972, validation accuracy of 0.9970, and test accuracy of 0.9880.

The Grade 4 GRU model used a similar architecture with a bidirectional GRU layer but with 128 neurons, a shorter window length of 25 samples, a dropout rate of 0.2, and a learning rate of 0.001. It achieved a training accuracy of 0.9989, validation accuracy of 0.9991, and test accuracy of 0.9730.

Among the classical machine learning models, normalized KNN achieved the highest validation accuracy (0.9424) and test accuracy (0.9099).

Table 1. Training, validation, and test accuracies for all classification algorithms, including normalized and not normalized neural networks and GRU models.

	KNN	Decision Tree	Norm. KNN	NN (50) Norm.	NN (50) Not Norm.	GRU (Grade 4)	GRU (Grade 5)
Training	—	—	—	0.9290	0.9309	0.9989	0.9972
Validation	0.9132	0.9085	0.9424	0.9066	0.9091	0.9991	0.9970
Test	0.8767	0.8799	0.9099	0.9180	0.9206	0.9730	0.9880

Figures 2 and 3 show the results of the confusion matrices for the Grade 4 and Grade 5 GRU models, respectively. The Grade 4 model demonstrates almost perfect classification across the four movement classes with only a few minor misclassifications. The Grade 5 model also achieves near perfect results despite the increased complexity due for the eight classes, it still maintained a clear diagonal pattern with very few misclassifications.

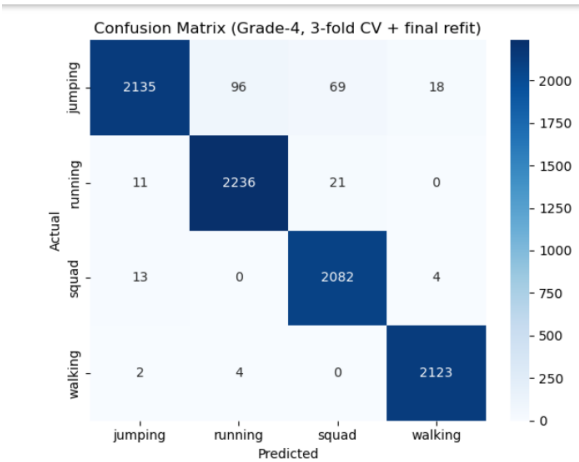


Figure 2. Confusion matrix for the GRU (Grade 4) model (participant 1).

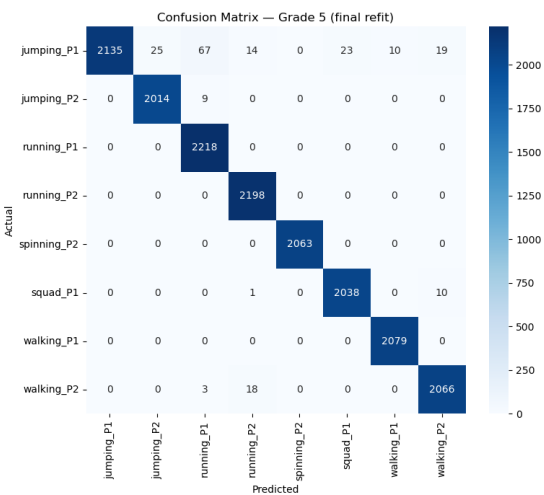


Figure 3. Confusion matrix for the GRU (Grade 5) model (participant 1 and 2).

The plot for the predicted activities for the 90-second combined file is shown for participant 1 in Figure 4, while the plot in Figure 5 shows the predicted activities for participant 2. The GRU model predictions for the 90-second recordings for Participant

1 (P1), correctly identifies jumping, running, and squats, while walking is only weakly detected and partly mixed with other movements. The movements are also predicted in the right order as they were recorded, starting with jumping followed by walking, squats and running. For Participant 2 (P2), the model captures the general activity sequence but shows more overlap and misclassifications, especially between movements and participants. For (P2) the spinning movement is the clearest one compared to the other prediction for (P2). The spinning movement appears in the end of participants 2 plot, which also matches with the actual recoding of the of the movements.

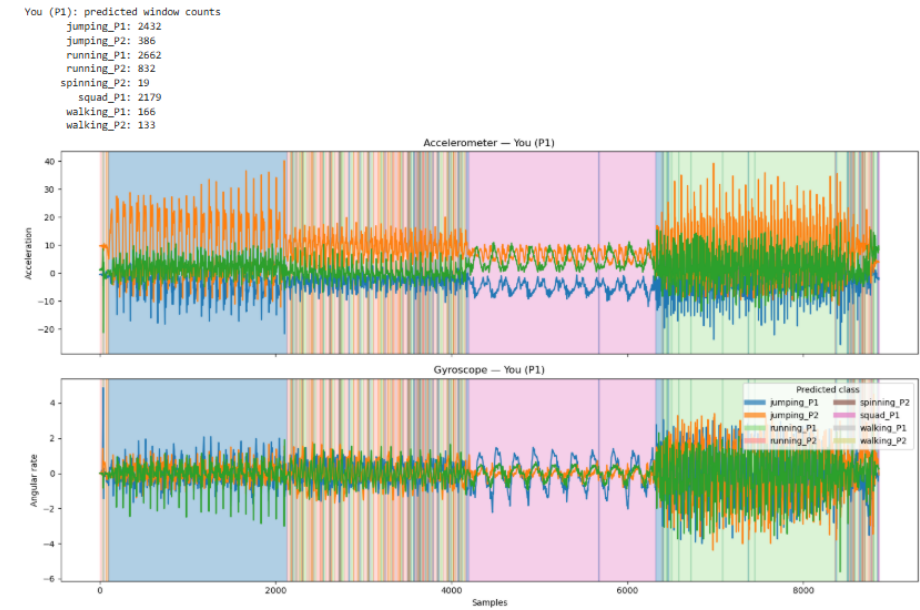


Figure 4. Predicted movement segments for participant 1, across accelerometer and gyroscope data.

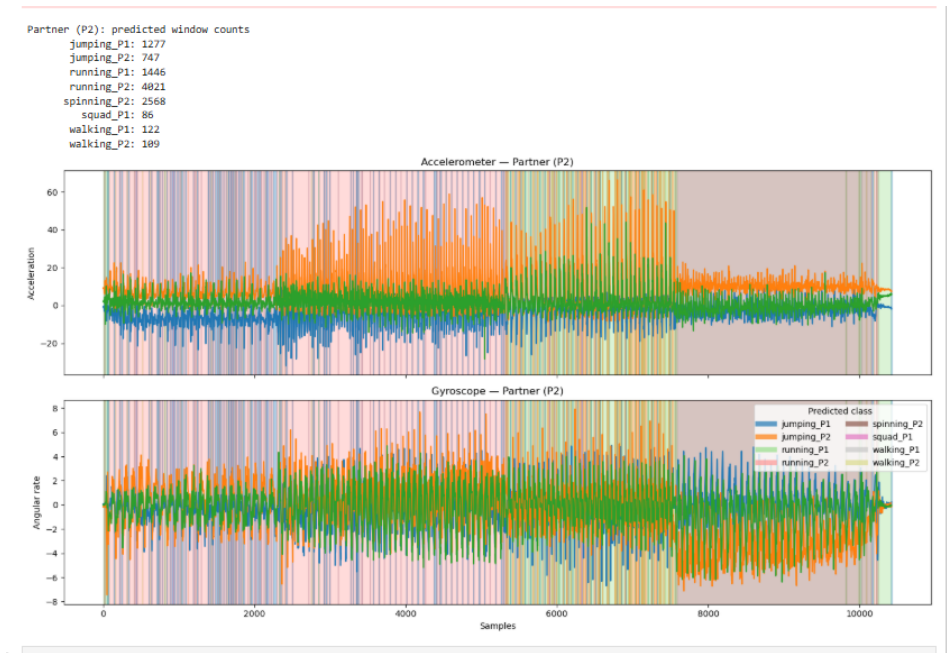


Figure 5. Predicted movement segments for participant 2, across accelerometer and gyroscope data.

4. Discussion

Comparison of all models shows differences in performance. The GRU models showed the best overall performance among all the approaches tested. The Grade 5 model achieved the highest test accuracy, successfully handling both movement and participant classification. A slight indication of overfitting was observed, as the training accuracy was marginally higher than the validation accuracy, suggesting that the model learned the training data slightly better than unseen data. Despite this recurrent neural network based on GRU is able of processing time dependent data for a more effective activity recognition than classical models and neural network. The GRU model for grade 4 also performed very well, correctly separating the four movements with only a few minor misclassifications. The small difference in accuracy between the two GRU models is explained by the increased complexity in Grade 5, which required the model to distinguish not only activities but also the person performing them.

The predicted activity segments from the 90-second combined recordings for participant 1 (P1), the GRU model correctly identified jumping, running, and squats, while walking was only weakly detected and occasionally mixed with other activities. The predicted segments also followed the correct order of recorded movements, indicating stable temporal recognition. For Participant 2 (P2), the overall movement sequence was captured, but the model showed more overlap and misclassifications between movements and participants. This suggests that while the model generalizes well to unseen data, participant specific movement patterns and sensor variation still affect prediction accuracy. The spinning activity in P2's data was the most clearly recognized, appearing at the end of the sequence in line with the actual recording.

The classification models, including KNN, normalized KNN, and Decision Tree, achieved lower overall accuracy compared to GRU models. Normalization improved the KNN results, demonstrating its sensitivity to feature scaling. The non-normalized neural network with 50 neurons outperformed the normalized version, indicating that raw sensor variations contained meaningful information that scaling partially removed.

Overall, the results confirm that GRU based recurrent models are the most suitable approach for classifying motion data captured by smartphone sensors. They effectively learn temporal patterns in accelerometer and gyroscope readings, though generalization across different individuals remains a challenge.